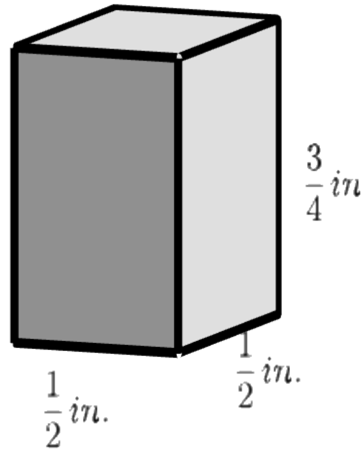
Grade 6 Accelerated
Unit 8
Lesson 2 Practice

Name___Answer Key______
Date____________________
Period___________________

A rectangular prism is shown with units measured in inches (in.).

In questions 1-4, find the volume using cubes with side lengths $\frac{1}{4}$ in.



1. Determine how many cubes will fit along the length, width, and height.

   $\frac{1}{2} \div \frac{1}{4} =$ ____2____ cubes will fit along the length.

   $\frac{1}{2} \div \frac{1}{4} =$ ____2____ cubes will fit along the width.

   $\frac{3}{4} \div \frac{1}{4} =$ ____3____ cubes will fit along the height.

2. A total of ____12____ cubes will fit within the rectangular prism.

3. The volume of each cube is $\frac{1}{64}$ in^3. Multiply the number of cubes needed to fill the prism by the volume of each cube to find the volume of the prism. Include units.

   $12\left(\frac{1}{64}\right) = \frac{12}{64} = \frac{3}{16}$ in.^{3}

4. Instead of using cubes, Kaia uses the formula $V = l \times w \times h$. Fill in the spaces below to complete her work. Include units.

   $V =$ __$\frac{1}{2}$__ $\times$ __$\frac{1}{2}$__ $\times$ __$\frac{3}{4}$__

   $V =$ __$\frac{3}{16}$__ in.^{3}

A miniature Rubik's cube has side lengths 0.8 inches (in.). The miniature cubes can be packed into a larger box with dimensions of 15 in. $\times$ 15 in. $\times$ 10 in.

5. How many miniature cubes can be packed into the larger box?

$15 \div 0.8 = 18.75$ $\quad$ $18.75 \times 12.5 \times 18.75 = 4{,}394.53125$

$10 \div 0.8 = 12.5$

$15 \div 0.8 = 18.75$

6. Use the number of miniature cubes that can be packed in the larger box to determine the volume of the box. Include units.

$4{,}394.53125(0.8^3) = 2{,}250$ in^3

Chera and Dylan are calculating the volume of a rectangular prism with the dimensions of 1.5 centimeters (cm) $\times$ 1.5 cm $\times$ 3 cm using different size cubes. Chera uses cubes with side lengths lengths $\frac{1}{4}$ cm, and Owen uses cubes with side lengths $\frac{1}{2}$ cm.

7. Who will need more cubes to fill the larger prism? Justify.

Chera, because her cubes have a smaller volume.

$\left(\frac{1}{4}\right)^3 = \frac{1}{64}$ cm^3 $< \left(\frac{1}{2}\right)^3 = \frac{1}{8}$ cm.^{3}

8. What is the volume of the rectangular prism? Include units.

$1.5 \times 1.5 \times 3 = 6.75$ cm.^{3}

9. If they had used a cube with side lengths $\frac{3}{4}$ cm instead, would they have found the same volume of the rectangular prism?

Yes

10. A sandbox in the shape of a rectangular prism is being filled with sand. The dimensions of the sandbox are 32.25 inches (in.) by 44.5 in. If the height of the sand is 15 in., what is the volume of the sand in cubic inches?

$32.25 \times 44.5 \times 15 = 21{,}526.875$ in.^{3}